Accounting Case Study

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QMB4900 Data Analytics Capstone

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Executive Summary

Super Shopper is experiencing inefficiencies in its ordering and in-store restocking procedures. Many items are critically understocked or exhibit a pattern of cyclical stockouts. The goal of this report will be to identify products that need to be reordered, calculate cost and revenue implications, and make recommendations to improve inventory and ordering process inefficiencies, from an Accounting Department perspective. Analysis techniques will be applied to Store 8 so that the process can be repeated for the remaining 9 locations.

Key Operations

Operation	Process
Inventory Tracking	Sales are tracked via barcode on store servers. Inventory updates are sent to the Warehouse weekly
Reordering Logic	Warehouse adds store-level inventory reports to master sheet with inventories from all 10 locations. Orders products below the reorder point
Pricing & Markups	Accounting updates retail pricing based on cost + markup formula
Ordering Process	Warehouse -> Accounting Approval -> Buyers -> Vendors
Order Cycle	Vendors ship next day

Fully Stocked vs Current Inventory

Category	Item	Fully Stocked	Current Inventory	Stock %	Reorder?
Kitchen Products	Dishwasher Soap	15	5	33%	Yes

Kitchen Products	Scouring Pads	20	0	0%	Yes
Kitchen Products	Window Cleaner	20	5	25%	Yes
Kitchen Products	Trash Bags	20	10	50%	Yes
Bathroom Products	Mouthwash	20	5	25%	Yes
Bathroom Products	Shampoo	20	5	25%	Yes
Bathroom Products	Hand Lotion	20	5	25%	Yes
OTC Medicines	Aspirin	20	5	25%	Yes
OTC Medicines	Vitamins	20	0	0%	Yes
OTC Medicines	Antacid	20	5	25%	Yes
OTC Medicines	Laxatives	10	0	0%	Yes
OTC Medicines	Motion Sickness	10	5	50%	Yes
Office Products	Pens	20	5	25%	Yes

Office Products	Ink Refills	10	0	0%	Yes
Office Products	Paper	15	0	0%	Yes

Store 8 currently has 15 products (60%) in need of restocking (current inventory at or below 50% fully stocked level). 12 of the 15 products requiring reorder (80%) have critically low inventory levels (current inventory at or below 25% fully stocked level). 5 Products are experiencing complete stockout.

Cost and Revenue Implications

A	B	C	D	E	F	G
Item	Units to Order	Cost/Unit	Total Cost	Retail/Unit	Retail Value	Expected Profit
Dishwasher Soap	10	$ 2.50	$ 25.00	$ 3.00	$ 30.00	$ 5.00
Scouring Pads	20	$ 2.00	$ 40.00	$ 2.40	$ 48.00	$ 8.00
Window Cleaner	15	$ 3.50	$ 52.50	$ 4.20	$ 63.00	$ 10.50
Trash Bags	10	$ 3.50	$ 35.00	$ 4.20	$ 42.00	$ 7.00
Mouthwash	15	$ 3.50	$ 52.50	$ 4.20	$ 63.00	$ 10.50
Shampoo	15	$ 5.50	$ 82.50	$ 6.60	$ 99.00	$ 16.50
Hand Lotion	15	$ 4.00	$ 60.00	$ 4.80	$ 72.00	$ 12.00
Asprin	15	$ 6.00	$ 90.00	$ 7.20	$ 108.00	$ 18.00
Vitamins	20	$ 5.50	$ 110.00	$ 6.60	$ 132.00	$ 22.00
Antacid	15	$ 5.00	$ 75.00	$ 6.00	$ 90.00	$ 15.00
Laxatives	10	$ 7.00	$ 70.00	$ 8.40	$ 84.00	$ 14.00
Motion Sickness	5	$ 6.50	$ 32.50	$ 7.80	$ 39.00	$ 6.50
Pens	15	$ 1.50	$ 22.50	$ 1.80	$ 27.00	$ 4.50
Ink Refills	10	$ 0.50	$ 5.00	$ 0.60	$ 6.00	$ 1.00
Paper	15	$ 4.50	$ 67.50	$ 5.40	$ 81.00	$ 13.50
Total	205	$ 61.00	820	$ 73.20	$ 984.00	$ 164.00

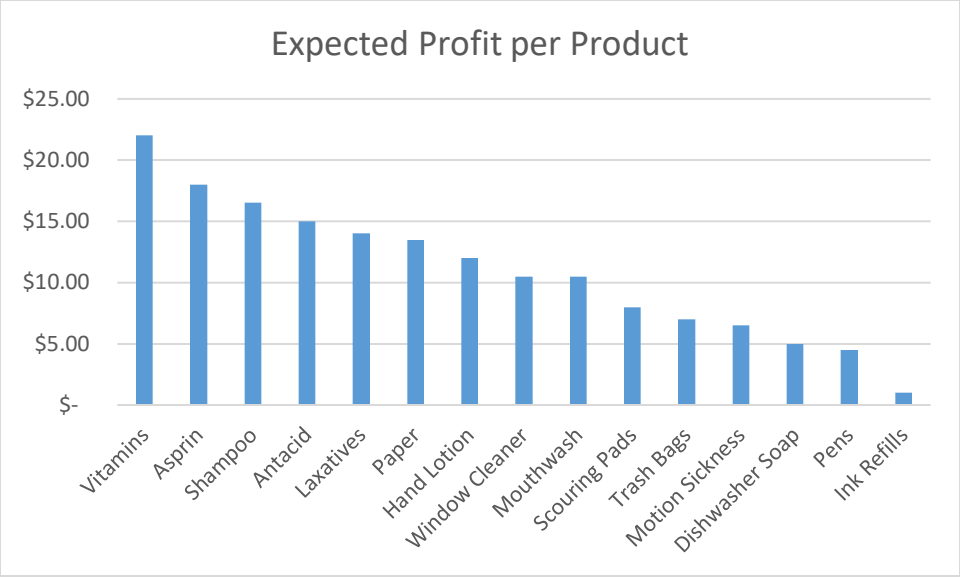
The table above displays the reordering needs for each product based on current inventory levels, and the related cost and revenue implications.

- Total Cost - $820
- Total Retail Value - $984
- Total Profit - $164

Visualizations & Recommendations

A	C	D	E	F	G	H	I	J
	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Present
	Inventory	Inventory	Inventory	Inventory	Inventory	Inventory	Inventory	Inventory
Kitchen Products								
Dish Soap	20	10	5	5	0	0	5	20
Dishwasher Soap	5	5	10	5	5	0	0	5
Scouring Pads	15	10	5	5	0	0	0	0
Window Cleaner	20	10	0	5	15	0	20	5
Paper Towels	15	5	20	10	10	5	0	20
Trash Bags	20	15	0	5	10	15	5	10
Bathroom Products								
Toothpaste	15	10	0	5	10	0	5	20
Mouthwash	10	10	5	0	5	10	10	5
Hand Soap	20	10	15	0	5	5	20	15
Shampoo	15	10	0	0	10	0	5	5
Hand Lotion	10	15	20	0	0	10	10	5
Toilet Paper	20	10	5	5	20	10	5	15
Cleaning Solvent	15	15	15	15	15	15	15	15
OTC Medicines								
Asprin	20	15	10	0	0	10	5	5
Vitamins	10	0	10	10	10	5	10	0
Antacid	15	5	5	0	5	0	10	5
Laxatives	10	5	10	10	5	0	0	0
Motion Sickness	10	5	5	0	5	5	10	5
Office Products								
Pens	10	10	5	15	5	10	10	5
Ink Refills	5	10	0	5	5	5	0	0
Pencils	20	20	20	20	20	20	20	20
Notebooks	15	10	0	10	15	10	5	15
Paper	15	15	10	5	15	10	5	0
Staplers	5	5	10	5	10	5	0	10
Paper Clips	5	5	0	5	0	5	10	10

The above heatmap details the inventory level for each product at Store 8 over the sample 8-week period. Cells in green represent a fully stocked item, cells in yellow represent a 50% stocked item, and red cells represent an item experiencing a stockout. Not only is the frequency of critically low inventory at the product level frequent and alarming, it is also clear to see that multiple products experience stockouts for consecutive weeks. Warehouse reorder points should be adjusted to represent inventory needs at the store level as the collective reorder points are leaving certain stores short.



This visualization shows the expected profit per product in need of reorder. Items with a higher expected profit should be prioritized should the Warehouse experience any ordering delays, or should Accounting discover any budgetary restraints.